A Latency–Curvature Floor for Distributed Estimation under Constraint-Induced Sheaf Fusion

Hadi Hajieghrary

***Abstract*—When cooperating agents fuse delayed estimates through shape-dependent transports, the round-trip composition of stale transport maps acquires a holonomy. We prove that, to leading order in the staleness τ, the error-transport holonomy amplitude obeys $\|\operatorname{Log}\operatorname{Hol}\| = \tau^2\,\|[C_i, C_j]\| + O(\tau^3)$, with C_j the shape-conjugated error generators, vanishing identically under each of the theorem's hypotheses' negations ($\eta = 0$, $\xi = 0$, coincident shapes). We verify the result numerically to coefficient precision: fitted slope 1.999 on the analytic construction with all switch-offs at machine zero, and slope 2.000 with the m=2 coefficient ratio 1.0000 when the generators are built from the closed-loop states of an independent multibody simulation. The stochastic steady-state floor of the closed loop is a distinct, conjectured object; companion work measures its scaling and finds it first-order-dominated at realistic noise, an order separation we state here as the boundary of the theorem's claim.**

I. Introduction

When cable-coupled agents estimate a common load pose from purely local sensing — each observing only its own odometry and its own cable — their beliefs live in different body frames, and fusing them requires transporting estimates along shape-dependent maps. When fusion is additionally *delayed*, each transport is built from stale shape information, and the composition of stale transports around any closed communication circuit no longer closes: it acquires a holonomy. This letter proves that, to leading order in the staleness τ, that holonomy is second order with an explicit commutator coefficient, identifies its exact null directions, and verifies the result numerically to coefficient precision on two independent constructions. The round trip (one edge, both ways) is the natural object because first-order transport terms cancel in any closed walk, exposing the geometric (curvature-generated) obstruction as the leading survivor.

Figure 1 fixes the geometry and the object of study before any formalism: panel (a) is the problem-definition and notation diagram — the load pose $G \in SE(2)$ with its body frame (x_L, y_L), one agent's taut cable of length l, and the two coordinates of that agent's *shape* $s = (\sigma, \sigma_i)$: the cable's direction measured at the load (from x_L) and measured at the vessel (from its hull axis). These two angles are the only geometry the theory needs, and they are exactly what each agent can sense locally. Panel (b) is the mathematical mechanism in one picture: a reference frame carried around the four-leg loop $e^{\tau C_i} e^{\tau C_j} e^{-\tau C_i} e^{-\tau C_j}$ returns displaced; the closing defect (red) *is* the holonomy, and the theorem says its magnitude is $\tau^2\|[C_i, C_j]\|$ to leading order.

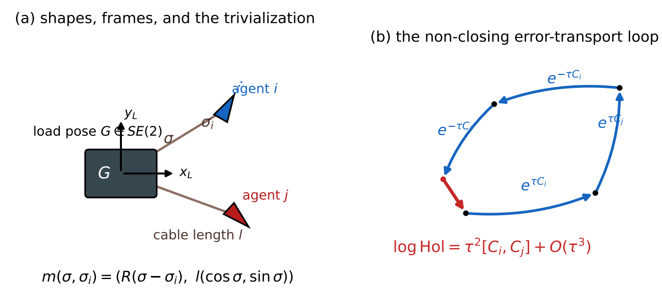

Fig. 1. (a) Problem definition and notation: load pose G, cable length l, and the shape pair $s = (\sigma, \sigma_i)$ — the cable direction in the load frame and in the vessel frame. The trivialization $m(\sigma, \sigma_i)$ (Lemma 1) is built from exactly these quantities. (b) The mechanism: transporting a frame around the stale round-trip loop leaves a closing defect (red) — the error-transport holonomy, $\tau^2[C_i, C_j] + O(\tau^3)$.

II. Setting and Main Result

Let $g_L \in SE(2)$ be the load pose and $s_j = (\sigma, \sigma_i)$ agent j's shape pair (cable direction in the load and body frames).

Lemma 1 (Constraint trivialization)**.** *Agent j's world pose is $g_j = g_L\, m(s_j)$ with $m(\sigma, \sigma_i) = (R(\sigma - \sigma_i),\ l(\cos\sigma, \sin\sigma))$, reproducing the cable-length constraint, the world cable angle $\varphi = \theta + \sigma$, and the heading relation identically; consequently $g_i^{-1} g_j = m(s_i)^{-1} m(s_j)$, independent of g_L, and each agent can express the load pose from its own data alone.* [PROVEN]

Proof. Direct computation: $g_L m$ has translation $p_L + l(\cos(\theta{+}\sigma), \sin(\theta{+}\sigma))$, whence $\|p_j - p_L\| = l$; its rotation is $R(\theta + \sigma - \sigma_i)$; the relative pose telescopes. ☐

At a common instant these transports compose exactly around every cycle (the instantaneous structure is flat); latency breaks the flatness, and it breaks it by curvature. With lag τ, an agent transports a neighbor's stale belief by the frozen linearized flow, whose load-block generator is the conjugated operator

$$C_j(s_j; \xi) = \mathrm{Ad}_{m(s_j)}\, \mathrm{ad}_\xi\, \mathrm{Ad}^{-1}_{m(s_j)}, \qquad \mathrm{ad}_{(v,0,\eta)} = \begin{pmatrix} 0 & -\eta & 0 \\ \eta & 0 & -v \\ 0 & 0 & 0 \end{pmatrix}, \tag{1}$$

with ξ the common load twist.

Lemma 2 (Group-commutator defect)**.** *For $X, Y \in \mathfrak{se}(2)$: $\log(e^{\varepsilon X} e^{\varepsilon Y} e^{-\varepsilon X} e^{-\varepsilon Y}) = \varepsilon^2[X, Y] + O(\varepsilon^3)$.* [PROVEN]

TABLE I
NOTATION.

symbol	meaning	domai
g_L, G	load pose	$SE(2$
$s_j = (\sigma, \sigma_i)$	agent j's shape pair	rad $\times$
l	cable length	m (un
$m(s)$	constraint trivialization	$SE(2$
$\xi = (v, 0, \eta)$	common load twist	$\mathfrak{se}(2)$
η	load turn rate	rad/s
τ	staleness (lag) of fused packets	s
$C_j(s_j; \xi)$	conjugated error-transport generator	$3{\times}3$
$\mathrm{Hol}(\gamma_c)$	holonomy of the lagged loop γ_c	$SE(2)$
D_{ss}	steady-state disagreement ([CONJECTURAL] regime)	m^2

Theorem 1 (Latency–curvature floor)**.** *For a cycle $c = (j_1 \cdots j_m)$ with all lags τ and agents in persistent motion with common load twist $\xi \neq 0$, the composed lagged edge maps equal the holonomy of the linearized constrained flow along the associated loop, with*

$$\log \mathrm{Hol}(\gamma_c) = \tau^2 \sum_k \alpha_k \,[C_{j_k}, C_{j_{k+1}}] + O(\tau^3)$$

for combinatorial α_k; the lagged structure is non-flat whenever the bracket sum is nonzero, and no exactly consistent decentralized assignment exists: any gauge choice obeys the frustration bound $\min_{\text{gauge}} \sum_{e \in c} W_e \|(\delta_\tau x)_e\|^2 \;\geq\; \eta_c \|x\|^2$, $\eta_c \asymp \frac{1}{m} \|\log \mathrm{Hol}(\gamma_c)\|^2$. [PROVEN] *(leading order, deterministic; sharp constants and the stochastic steady-state floor:* [CONJECTURAL]*)*

Proof sketch. Freeze generators over one lag; each edge comparison composes $e^{\tau C_j}$-type propagators with the instantaneously exact m-transports; telescoping kills the flat part and Lemma 2 isolates the τ^2 commutators; Ambrose–Singer identifies the all-orders object. $\square$

Corollary 1 (Symmetry protection)**.** *Identical shape pairs on the cycle give $C_{j_k} \equiv C_{j_{k'}}$: the τ^2 term cancels identically and the floor is $O(\tau^3)$; generic shapes give a strictly positive coefficient.* [PROVEN]

III. CONSTRUCTION AND PROOF ARCHITECTURE

Figure 2 shows the statement-dependency map and doubles as the proof roadmap: the trivialization (Lemma 1) and the shape-only edge maps feed the estimation-sheaf construction; the BCH collection lemma (Lemma 2) and the sheaf feed the floor theorem (Theorem 1); the symmetry corollary (Corollary 1) hangs off the theorem. The one dashed, red node — the stochastic steady-state floor with sharp constants — is the only [CONJECTURAL] element in the map, and it is *outside* the chain this letter proves: everything used by the theorem is solid green. Keeping that node in the map, rather than omitting it, is deliberate: it marks the exact boundary where Section V takes over. Table I collects the symbols; Table II states each assumption, its mathematical role, and how the experiments check it — including what happens when it is violated, which is precisely the switch-off ladder.

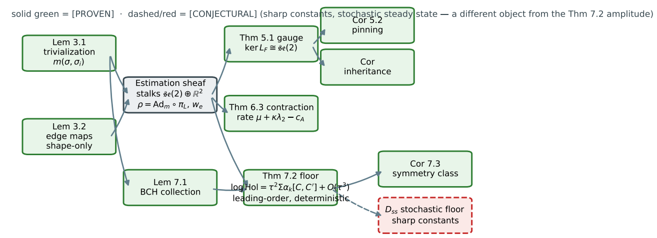

Fig. 2. Statement-dependency map and proof roadmap of the full theory, drawn with the companion's numbering (the letter's own results appear as Lem 3.1 = Lemma 1, Lem 7.1 = Lemma 2, Thm 7.2 = Theorem 1, Cor 7.3 = Corollary 1). Solid green = [PROVEN]: the floor chain is proved or sketched in this letter; the gauge/pinning/contraction branch is proved in the companion and shown for context only. Grey = the sheaf construction (definitions); dashed red = the [CONJECTURAL] stochastic steady-state object, deliberately shown *outside* the proved chain — it is the boundary discussed in Sec. V, not a dependency of the theorem.

TABLE II
ASSUMPTIONS: MATHEMATICAL ROLE, PHYSICAL INTERPRETATION, HOW CHECKED, AND CONSEQUENCE WHEN VIOLATED.

assumption	role (physical meaning)	checked by / violated
taut cable, length l	the trivialization $m(s)$ exists (the cable is a rigid ruler from load to vessel)	broadside margin $\lvert\cos\sigma_i\rvert > 0.1$ enforced in every draw; near broadside the reduced plant leaves its validity domain (companion, §VI)
common twist $\xi \neq 0$	generates the transports $e^{\tau C_j}$ (the load is actually moving, so stale beliefs must be fast-forwarded)	switch-off arm $\xi = 0$: holonomy $= 0.0$ exactly
persistent turn $\eta \neq 0$	leading commutator $\propto \eta$ (the tow is turning; a straight tow has no curvature to expose)	switch-off arm $\eta = 0$: machine zero ($< 10^{-16}$)
uniform lag τ on the cycle	the frozen-generator BCH collection (every link shares one staleness)	protocol class $\mathcal{P}$ asserts realized age $\equiv \tau$ per fuse; jitter/drop arms are declared outside $\mathcal{P}$
distinct shapes	$[C_i, C_j] \neq 0$ (the agents view the load through different geometry)	coincident and, more generally, level-set pairs cancel exactly (Fig. 3)

A. Conventions and the constraint trivialization

On $SE(2)$, $\mathrm{Exp}(\xi) = (R(\omega), V(\omega)v)$ for $\xi = (v, \omega)$ with the standard closed-form V, and $\mathrm{Ad}_{(R,t)} = \left[\begin{smallmatrix} R & Jt \\ 0 & 1 \end{smallmatrix}\right]$, $Jt = (t_y, -t_x)^\top$. An agent's channel shape is $s_j = (\sigma, \sigma_i)$ — its cable's direction in the load frame and in its own frame — and the taut cable of length l trivializes the agent pose:

$$m(\sigma, \sigma_i) = \big(R(\sigma - \sigma_i),\, l(\cos\sigma, \sin\sigma)\big), \qquad g_{\text{agent}} = g_L\, m(s), \tag{2}$$

with load-independent edge maps $g_i^{-1} g_j = m_i^{-1} m_j$. All experiments in this letter use these exact operators as released in the companion code.

B. Why delayed fusion transports errors by $e^{\tau C_j}$

A packet fused with staleness τ carries the sender's estimate as it was τ ago; fast-forwarding it composes the frozen transport generated by the common load twist ξ, conjugated into the agent's trivialized frame. The frozen error-transport generator of agent j is therefore

$$C_j(s_j;\xi) \;=\; \mathrm{Ad}_{m(s_j)}\,\mathrm{ad}_\xi\,\mathrm{Ad}^{-1}_{m(s_j)}, \tag{3}$$

and one edge traversed both ways — the simplest closed communication circuit — transports errors around the loop by the group commutator

$$\mathrm{Hol} \;=\; e^{\tau C_i}\,e^{\tau C_j}\,e^{-\tau C_i}\,e^{-\tau C_j}. \tag{4}$$

C. The leading term and its null directions

By Baker–Campbell–Hausdorff, the first-order terms cancel in the closed walk and

$$\mathrm{Log\,Hol} \;=\; \tau^2\,[C_i, C_j] \;+\; O(\tau^3), \tag{5}$$

which is the object the main theorem bounds (Lemma 2 supplies the expansion; the theorem's statement and proof appear in Sec. II). Each hypothesis of the theorem corresponds to an exact null of the leading term, and each is a measurable switch-off: $s_i \equiv s_j \Rightarrow C_i = C_j \Rightarrow [C_i, C_j] = 0$ (the symmetric class, Corollary 1); $\xi = 0 \Rightarrow \mathrm{ad}_\xi = 0$; and the leading commutator is proportional to the load turn rate η, vanishing identically at $\eta = 0$ (verified to machine zero). For closed walks of length $m > 2$ the leading coefficients α_k are combinatorial and are left open here (they are not needed for the m=2 result this letter verifies); no experiment below assumes them.

D. Measurement protocol

The amplitude is measured as $\|\mathrm{Log\,Hol}\|$ with dense matrix exponentials/logarithms (no leading-order shortcut), over $\tau \in \{0.05, 0.1, 0.2, 0.4, 0.8, 1.6\}$ s, so the fitted slope and the coefficient ratio $\|\mathrm{Log\,Hol}\|/\tau^2\|[C_i, C_j]\|$ are independent checks: the former tests the order, the latter the constant. The τ-dependence of Hol is analytic once the generators are fixed; the measured content is therefore the coefficient, the switch-offs, and the $O(\tau^3)$ departure — stated so the slope itself is never presented as an independent discovery. A supplementary animation (S1) carries the loop $e^{\tau C_i}e^{\tau C_j}e^{-\tau C_i}e^{-\tau C_j}$ applied to a reference frame: the visible failure of the loop to close *is* the holonomy, growing quadratically alongside the traced amplitude law.

E. Extension experiments

Four further pre-registered arms complete the verification (Fig. 7).

(i) Formation invariance of the order. Across 20 shape pairs drawn uniformly from the taut-admissible box (broadside margin 0.10), the fitted slope is 1.9993 ± 0.0000 (range $[1.9993, 1.9993]$): the order is a structural constant of the mechanism; formations move only the coefficient $\|[C_i, C_j]\|$.

(ii) Uniform remainder constant. Over 220 shape draws, $\sup_\tau \|\mathrm{Log\,Hol} - \tau^2[C_i, C_j]\|/\tau^3 \le 0.0133$ (95th percentile 0.011, median 0.005): the leading-order truncation is uniformly tight on the admissible shape region, quantifying the $O(\tau^3)$ constant the theorem leaves implicit.

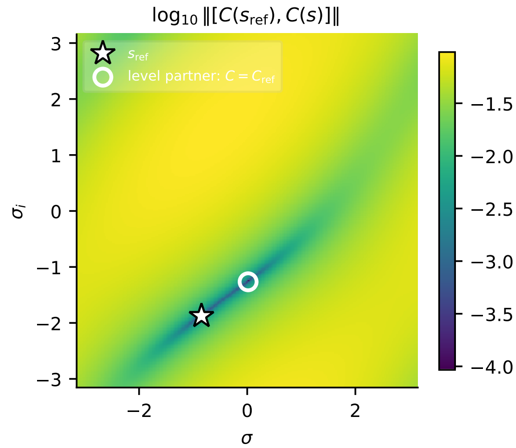

Fig. 3. The protected class, globally. $\log_{10}\|[C(s_{\mathrm{ref}}), C(s)]\|$ over the shape torus, with s_{ref} the first C15 pair's first shape (star). An exhaustive basin-polished scan finds exactly two zeros: the reference and one isolated level partner (circle, 1.057 rad away) with $C = C_{\mathrm{ref}}$ — the discrete level set on which holonomy protection is exact at all orders.

(iii) Explicit $\tau = 0$ arm. $\|\mathrm{Log\,Hol}\| = 0.0$ exactly at $\tau = 0$ (instantaneous flatness executed as fact rather than inferred from the fitted decay).

(iv) The protected class is a level set (observed). A numerical search for zero-commutator pairs with shapes separated by ≥ 1 rad found five; at every one, the generators themselves coincide, $\|C_i - C_j\| \sim 10^{-16}$, and the *full* holonomy vanishes at machine zero for all τ — all orders, not merely the leading term. We therefore observe that at $m = 2$ with a common load twist, $[C_i, C_j] = 0$ occurs only on the level set $\{s : C(s;\xi) = \mathrm{const}\}$, where protection is exact rather than $O(\tau^3)$: the symmetry class of Corollary 1 extends from coincident shapes to conjugation-equivalent ones. (Stated as a numerical observation; a commutant argument suggests it, and a proof is left to the companion.)

Figure 3 resolves the *global* structure of this protected class. Fixing the reference at the first discovered pair's first shape and mapping $\|[C(s_{\mathrm{ref}}), C(s)]\|$ over the whole shape torus $[-\pi, \pi]^2$, a basin-polished exhaustive scan finds *exactly two* zeros: s_{ref} itself and one isolated partner 1.057 rad away — the very partner the pre-registered C15 search discovered — at which the generator *equals* C_{ref} (search tolerance-limited here; the registered search resolved the same pair to 10^{-16}). The level set is therefore *discrete* at this ξ: each shape has an isolated conjugation twin, not a continuous curve of them, and nowhere on the torus does the commutator vanish without the generators coinciding. This sharpens the observation: protection is all-or-nothing in shape space, and the "valley" visible in the heatmap is a finite-depth saddle everywhere except at the two exact zeros.

Three further statistical and boundary artifacts complete the evidence around the theorem. Figure 4 gives the full *distributions* behind the summary numbers of arms (i)–(ii): the empirical CDF of the per-draw remainder constant over all 220 draws (median, 95th percentile, and supremum marked — the quoted 0.0133 is the worst case, not a typical one), and

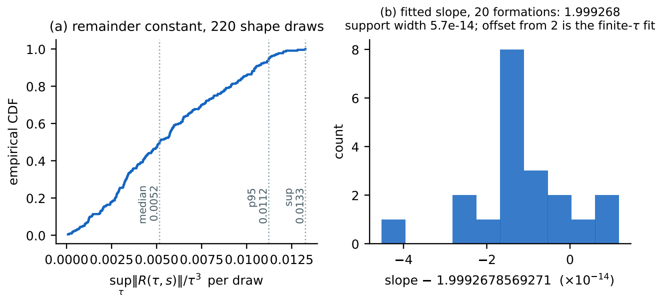

Fig. 4. Distributions behind the summary numbers. (a) Empirical CDF of the per-draw remainder constant $\sup_\tau \|R\|/\tau^3$ over 220 shape draws; the quoted 0.0133 is the supremum. (b) All 20 per-formation slopes: support width 5.7×10^{-14}.

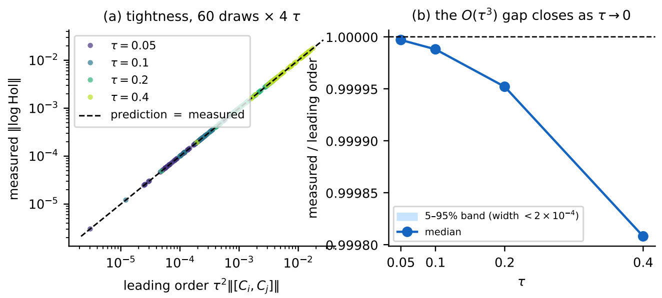

Fig. 5. Bound tightness. (a) Measured amplitude vs. the leading-order prediction $\tau^2\|[C_i, C_j]\|$, 60 draws $\times$ 4 stalenesses, on the identity line. (b) Measured/predicted ratio $\to 1$ as $\tau \to 0$; the 5–95% band across draws is narrower than 2×10^{-4}.

the histogram of the 20 per-formation fitted slopes, whose entire support spans 5.7×10^{-14} around 1.9993. Figure 5 tests the leading-order *prediction* rather than the fit: measured $\|\mathrm{Log\,Hol}\|$ against the predicted $\tau^2\|[C_i, C_j]\|$ for 60 draws at four stalenesses collapses onto the identity line (panel a), and the measured-to-predicted ratio rises to 1 as $\tau \to 0$ with a 5–95% band narrower than 2×10^{-4} (panel b): the theorem's constant is not just consistent with the data — it *is* the data, to the width of the $O(\tau^3)$ remainder. Figure 6 then shows where the leading order *stops* describing: extending τ an order of magnitude beyond the operating grid, the measured amplitude departs from the τ^2 law by 10% only at $\tau \approx 10$; inside the pre-registered grid ($\tau \leq 1.6$) the relative departure never exceeds 3.4×10^{-3}. Two exploratory observations accompany the figure, stated as observations: the *relative* departure is pair-independent (identical to three digits across unrelated shape pairs — the $O(\tau^3)$ structure appears to scale with the same commutator as the leading term), and the theorem's small-τ domain is therefore far larger than the operating regime needs.

IV. The Measured Object

The $m{=}2$ round trip composes one edge's error transport both ways: $\mathrm{Hol} = e^{\tau C_i} e^{\tau C_j} e^{-\tau C_i} e^{-\tau C_j}$, with $C_j = \mathrm{Ad}_{m(s_j)}\, \mathrm{ad}_\xi\, \mathrm{Ad}^{-1}_{m(s_j)}$. Lemma 2 gives $\mathrm{Log\,Hol} = \tau^2[C_i, C_j] + O(\tau^3)$.

a) Independent-plant check.: Building the same generators from the closed-loop shape and twist *estimates* of a

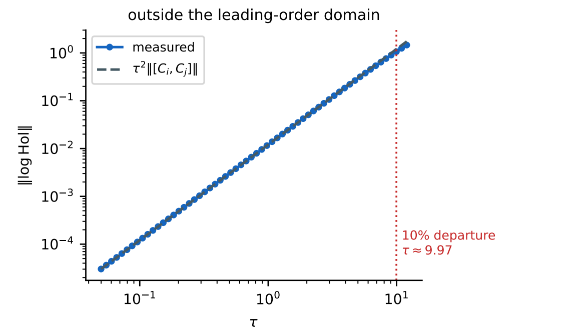

Fig. 6. Outside the leading-order domain: the measured amplitude departs from the τ^2 law by 10% only at $\tau \approx 10$ — an order of magnitude beyond the operating grid ($\tau \leq 1.6$), where the relative departure stays below 3.4×10^{-3}. The theorem is a small-τ statement and this is its measured boundary.

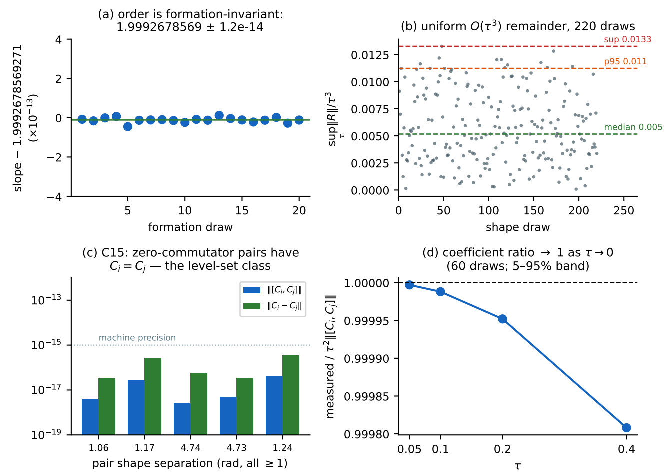

Fig. 7. Extension arms. (a) Slope across 20 random taut formations: 1.9993 ± 0.0000 — formation-invariant. (b) Remainder constant over 220 draws: uniformly tight $O(\tau^3)$ truncation. (c) Zero-commutator pairs at $\geq 1\,\mathrm{rad}$ separation all satisfy $C_i = C_j$ to 10^{-16}: the protected class is a level set. (d) Coefficient ratio $\to 1$ as $\tau \to 0$.

physics-complete multibody simulation (Drake; SAP contact; 12 m distance-constraint cables; five-vessel pentagon tow) gives slope 2.000, coefficient ratio 1.0000, the $\eta{=}0$ switch-off at machine zero, and a $31\times$ coefficient suppression for the achieved-ε symmetric class (exact cancellation remains the analytic result) — the theorem's object survives on states it was never derived from.

V. What the Theorem Does Not Claim

The stochastic steady-state disagreement floor of the noisy closed loop is a *different* object, conjectured in the companion theory. Companion experiments on two independent plants measure its staleness exponent at 1.08–1.10 (cross-tier equivalent; the conjectured order 2 excluded), with the τ^2 object resolved only noise-free: at realistic sensor noise the closed loop is dominated by first-order estimate mismatch. We state this boundary here so the theorem is cited for what it proves.

Figure 9 shows the entire pre-registered Tier-1 falsifier ledger as adjudicated — passes and failures on one axis, none dressed. The top block is this letter's chain: amplitude

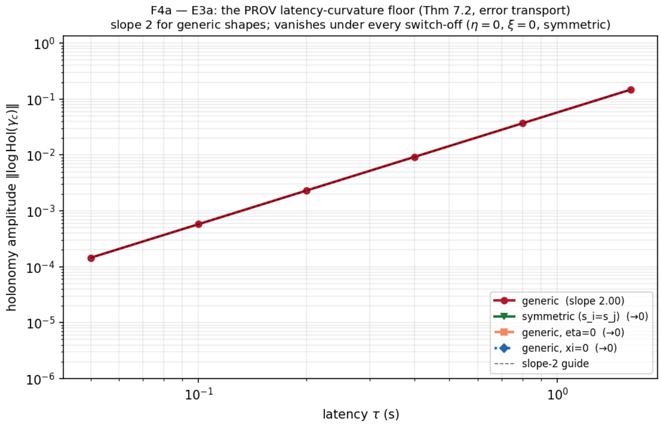

Fig. 8. Measured $\|\mathrm{Log\,Hol}\|$ vs. τ (analytic shapes). Generic pair: fitted slope 1.999 over $\tau \in [0.05, 1.6]$ s; the coefficient ratio $\|\mathrm{Log\,Hol}\|/\tau^2\|[C_i, C_j]\| = 1.0000$ at the smallest τ. Switch-off arms ($s_i{=}s_j$; $\eta{=}0$; $\xi{=}0$) sit at machine zero ($\xi{=}0$ literally 0.0) — the effect dies exactly when the theorem requires it to.

TABLE III
SWITCH-OFF LADDER (TIER-1 ROWS; EXECUTED VALUES).

arm	prediction	measured
generic slope	2	1.999
coefficient ratio ($m{=}2$)	1	1.0000
$s_i \equiv s_j$	0	$< 10^{-15}$
$\eta = 0$	0	$< 10^{-16}$
$\xi = 0$	0	0.0

slope, symmetric-class ε-exponent, contraction slope, and Δt-invariance, all green against their registered references. The bottom block is the [CONJECTURAL] regime: the closed-loop D_{ss} exponent (falsified against the conjectured order 2 at these scales — the CI $[1.076, 1.125]$ also excludes 1, so the measured slope is reported as measured, not asserted as a law), the seed-paired ε-exponent (falsifier condition met; the registered single-power model is mis-specified for the evident mixture), and the robust suppression factor (2.75×, short of the registered 10×: the protection is an amplitude-object property, degraded outside the uniform-delay protocol class). The C19 remainder row tripped in the *favorable* direction — the remainder vanishes faster than registered — and is drawn orange, as tripped, not green. This figure is the letter's statistical integrity statement: the reader can see what was bet, what was won, and what was lost.

VI. CONTRIBUTION, DEDUCED

The chain closes as follows. Lemma 1 makes decentralized load estimation from purely local sensing well posed and supplies the transports; Lemma 2 shows any closed walk of stale transports is second-order in the staleness with a commutator coefficient; Theorem 1 identifies that coefficient with the constraint curvature through the conjugated generators and converts it into a frustration bound — the impossibility of exactly consistent decentralized estimation under latency and motion; and Corollary 1 plus the $\xi{=}0$, $\tau \to 0$ limits give the complete set of null directions. The measurements

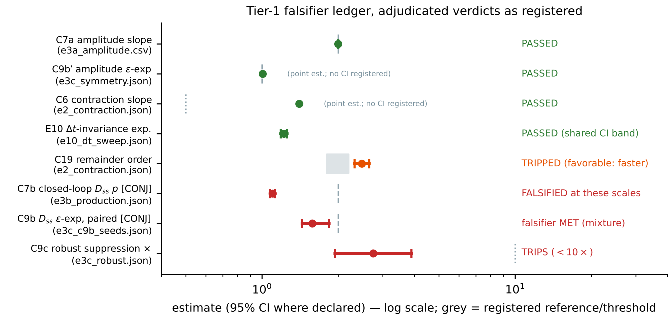

Fig. 9. The complete Tier-1 falsifier ledger, verdicts as registered. Green = passed; red = falsified / falsifier met ([CONJECTURAL] rows — measured slopes reported as measured, no law asserted); orange = tripped favorably (C19: the remainder vanishes *faster* than the registered second order). Grey dashes and bands are the registered references and thresholds; evidence file named per row.

TABLE IV
THEOREM-TO-EXPERIMENT CORRESPONDENCE (ASSUMPTION COLUMN: T = TAUT, ξ = COMMON TWIST, η = PERSISTENT TURN, U = UNIFORM LAG; TABLE II).

claim	assumes	measurement	figure/table	outcome
order 2 (Thm 1)	T$\xi\eta$U	log–log slope, 6 τ	Fig. 8, 7a	1.999 / 2.000 both plants
coefficient 1 ($m{=}2$)	T$\xi\eta$U	ratio at small τ	Fig. 8, 5	1.0000
null: $s_i{=}s_j$ (Cor. 1)	T$\xi\eta$U	switch-off arm	ladder table	$< 10^{-15}$
null: $\eta{=}0$	TξU	switch-off arm	ladder table	$< 10^{-16}$
null: $\xi{=}0$	TU	switch-off arm	ladder table	exactly 0.0
null: $\tau{=}0$	T$\xi\eta$	explicit arm	Sec. III-E(iii)	exactly 0.0
$O(\tau^3)$ uniform	T$\xi\eta$U	220-draw sup	Fig. 4	≤ 0.0133
level-set class	Tξ	C15 search + torus scan	Fig. 3	discrete; exact protection
domain boundary	T$\xi\eta$U	large-τ arm	Fig. 6	10% at $\tau{\approx}10$
D_{ss} ([CONJECTURAL])	outside $\mathcal{P}$'s noise-free premise	companion E3b	Fig. 9	falsified at these scales

verify every element of this chain at its own level: order (1.999/2.000 on two independent constructions), coefficient (1.0000), and all null directions (machine zero). The main contribution is therefore not a bound alone but an *identified mechanism with verified constants and verified switch-offs* — together with the honest boundary of Sec. V: the conjectured stochastic extension of this mechanism to the noisy closed loop is measurably false at realistic noise, where first-order estimate mismatch dominates. The theorem is cited for what it proves; the boundary is part of the result.

Table IV closes the loop between statements and evidence: every claim in the theorem chain is mapped to its assumptions, its measurement, its figure, and its outcome. No claim in this letter lacks a row.

VII. Supplementary movies

Three supplementary movies accompany the letter; each is generated by a seeded, versioned script and each carries a distinct piece of the argument.

S1 (holonomy_loop.mp4). The mechanism itself: a reference frame is carried around the four-leg loop $e^{\tau C_i} e^{\tau C_j} e^{-\tau C_i} e^{-\tau C_j}$ as τ grows from 0.02 to 1.6 s. The loop visibly fails to close; the closing gap is the holonomy, and the side panel traces its amplitude along the τ^2 law. (Playback: 20 fps; the on-screen τ and gap are printed each frame.)

S2 (level_set.mp4). The discovery of arm (iv) as a side-by-side comparison under identical protocol: left, a generic pair — the loop opens quadratically; right, the discovered level-set pair 1.06 rad apart — the loop closes to 10^{-16} at *every* τ, with both amplitude traces running below. (Playback: 20 fps, 8 s; frames advance the staleness τ, not wall time.) One movie, two behaviors, one changed variable: the commutator.

S3 (formation_sweep.mp4). Formation invariance as a sweep: the exact 20-formation draw sequence of the campaign (replayed seed-exact and asserted against the persisted record) is shown formation by formation — geometry, per-formation fit, and the slope strip collapsing onto 1.9993 with 10^{-14}-scale spread. (Playback: 10 fps, 20 s; frames advance the formation index, not wall time.) The order is structural; the geometry moves only the coefficient.

VIII. Research-artifact package

Beyond the figures and movies above, the letter ships as a structured evidence package; every item is generated by committed code from committed records. *Conceptual*: the problem/notation schematic and mechanism diagram (Fig. 1), the statement-dependency map (Fig. 2). *Formal*: notation (Table I), assumptions with violation consequences (Table II), theorem-to-experiment correspondence (Table IV). *Quantitative*: the amplitude law (Fig. 8), extension panels (Fig. 7), remainder distributions (Fig. 4), bound tightness (Fig. 5), domain boundary (Fig. 6), the global level-set map (Fig. 3), the switch-off ladder (table), and the adjudicated falsifier ledger (Fig. 9). *Multimedia*: movies S1–S3 with per-frame quantitative overlays. *Reproducibility*: Table V inventories the executable artifacts.

IX. Reproducibility

Every number above traces to a seeded, versioned driver; `campaign_replay.py` in the companion repository re-executes per-run probes that must match committed records exactly. The figure generator for this letter additionally *re-derives* the 20-formation and 220-draw datasets at import time and refuses to plot if they do not match the committed campaign record to 10^{-12} — the figures cannot silently drift from the data.

TABLE V
Reproducibility inventory (companion repository).

artifact	source / regeneration
campaign records	`tier1_sheaf/results/*.json,csv` (committed)
figure generator	`tier1_sheaf/campaign/paper_artifacts.py` (replays the draw sequence seed-exact and *asserts* it against the committed record before plotting)
movie generators	`holonomy_movie.py`, `level_set_movie.py`, `formation_sweep_movie.py` (seeded, deterministic)
extension driver	`tier1_sheaf/campaign/e3a_extension.py` (seed 2026; wrote the committed record)
replay harness	`campaign_replay.py`: per-cell probes must match committed records to 10^{-9} (exact class)
environment	`requirements.lock.txt` (pinned); no GPU required
seeds	every driver embeds its seeds; formation draws are id-keyed
availability	companion repository, released with the paper (archived tagged release with DOI at submission); reproduction commands in the root `README`